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GrowTH

A Web App for parents to track their child's growth

Note: *FR-23 requires all clinical or quasi-clinical results to be presented in plain, parent-friendly language. It appears across every scenario because each one ends with the app returning a result (growth summary, screening result, or bone age explanation) that must be understandable to a non-medical parent — making it a requirement shared by all personas.*

Persona 1 : Nan *FR-1,2,3,4,6,7,9,10,11,22*

Photo	
Name	แนน (Nan)
Age	32
Role	Primary user
Occupation	Office Employee
Background	Married, one daughter aged 2 years old
Tech Savvy	Intermediate, comfortable with everyday apps (social media, banking apps, food delivery apps)
Problem	• Unable to interpret the growth charts in the Child Health Record Book. • Often forgets to bring the Child Health Record Book. • Does not fully trust BMI or percentile values found on general websites or the internet.
Motivation	• Nan noticed that her daughter appeared smaller than other children of the same age, which made her anxious about whether this was a sign of a growth problem. Not wanting to wait weeks until the next scheduled doctor's visit to find out, she felt an urgent need to verify her child's growth status sooner rather than later.

Expectation	Wants an easy-to-understand system for tracking growth data and early detection of potential abnormalities.
Goal	- Wants to know whether their child's growth is developing normally without having to wait for the next doctor's appointment. - Wants to keep an easy-to-understand visual record of their child's growth history. - Wants to detect potential growth abnormalities at an early stage.

Scenario A+B: *FR-1,2,3,4,6,7,9,10,11,22*

Nan is concerned that her 2-year-old daughter appears smaller than other children her age. Feeling anxious about her child's development, she opens the traditional paper growth record booklet but finds the charts and graphs extremely confusing and difficult to interpret, leaving her unsure if her child is growing normally. Furthermore, her records are incomplete because she frequently forgets the booklet at home—especially when going to see the doctor. Frustrated by the confusing charts and missing data, Nan searches for information online, only to find an overwhelming amount of data and conflicting numbers, making her deeply distrust the unverified BMI values and percentile stats found on the internet. Realizing she would have to wait until the next doctor's appointment just to know the truth, Nan feels that waiting is too risky, believing that catching any growth delays early is much better for her child. Seeking a solution, she remembers a recommendation for a web application and decides to use it. That evening, she opens GrowTH on her smartphone (*FR-22*), signs up for an account (*FR-1*) and reviews and accepts the terms of service and privacy notice (*FR-2*) before proceeding. She then enters her child's profile details (*FR-4*) and recent measurements (*FR-6*), and instantly receives the results (*FR-7*) also views a line graph plotting her daughter's growth trend against standard percentile curves (*FR-9*), and browses through past growth entries to see how her daughter has been developing over time (*FR-11*) accompanied by clear, simple, and plain-language explanations (*FR-10*) that immediately relieve her anxiety and show her child's exact growth status. A few days later, during her lunch break at the office, she logs back in (*FR-3*) from her desktop computer and finds the same information displayed clearly, with the layout automatically adjusting to the larger screen (*FR-22*).

Persona 2 : Somchai *FR-3,5,6,7,8,12,13,14, 20,21*

Photo	
Name	สมชาย (Somchai)

Age	40
Role	Primary user
Occupation	Small Business Owner
Background	• Married, two children : a daughter aged 9 and a son aged 5. • Manages growth and development tracking for both children, who are at different developmental stages.
Tech Savvy	• Moderate to high — runs his own business and regularly uses digital tools for work (accounting apps, LINE OA, online banking), comfortable navigating new apps independently.
Problem	• Doesn't know who to consult when notices a potential concern with the child. • Information found online is inconsistent and conflicting, making it hard to judge whether the situation is urgent or normal. Also finds it inconvenient to track two children's records separately using paper health books.
Motivation	• Feels uncertain and slightly anxious noticing early physical changes in the daughter, but doesn't know who to ask. Online information is often contradictory, which increases anxiety rather than resolving it. With two children at different stages, wants one reliable tool that let users manage both without switching between accounts, and a trustworthy first step before committing time to a doctor's appointment.
Expectation	• A guided, structured way to check whether an observation is within normal range, with a clear, plain-language recommendation on whether to see a doctor now or monitor further plus trustworthy educational content to reduce uncertainty. Also expects to easily switch between two children's profiles without confusion.
Goal	• Wants to know whether something noticed about the daughter's development is actually unusual before deciding to see a doctor, while also keeping a simple track of the younger son's growth and easily switching between two children's profiles.

(Scenario C) : FR-3,12,13,14

Somchai notices a physical change in his 9-year-old daughter but is unsure if it's normal and doesn't know who to ask. He opens GrowTand logs into his account (*FR-3*), selects her profile, and completes the guided puberty screening questionnaire (*FR-12*). The web returns a plain-language result — clearly labeled as a screening aid, not a diagnosis — advising whether to consult a doctor now or continue monitoring (*FR-13*). Before deciding what to do next, he

checks her past screening history to see if any similar signs were noted in previous assessments (FR-14).

(Scenario F) : FR-20,21

Still in the app, he taps “Read more” (FR-20) on the result to better understand the topic before a possible doctor’s visit, browsing (FR-21) to the puberty development category and reading a short, plain-language article.

(Scenario E) : FR-5,6,7,8

Before closing the app, he remembers he still needs to log his 5-year-old son’s growth data from a clinic visit earlier that day. He switches profiles (FR-5) from the dashboard, enters the new height/weight including date of measurement (FR-6) for his son, and the app calculates and display the percentile and standard deviation score (SDS) for height-for-age and weight-for-age (FR-7) as well as the BMI and BMI-for-age result (FR-8), since his son has just turned 5 — keeping both children’s records completely separate.

Persona 3 : Mol FR-3,15,16,17,18,19,24

Photo	
Name	มล (Mol)
Age	38
Role	Primary user
Occupation	Shift Nurse
Background	• Single mother, one son aged 13 years old.
Tech Savvy	• High — works in healthcare, familiar with medical terminology and comfortable using digital health tools and hospital systems.
Problem	• Appointment time with specialists is limited, making her feel unable to fully explain observations. Also concerned about the privacy and security of uploading the son's X-ray image.
Motivation	• As a nurse, understands that specialist appointment time is short and valuable. Wants to walk in prepared with objective information rather than relying only on verbal observations, so the consultation time is used efficiently. Also cautious about

	the privacy of the son's medical images.
Expectation	• A secure, straightforward way to upload an X-ray image and receive a clear result with an explanation, presented in a way that supports an upcoming conversation with the specialist, with confidence that the image and data are kept private and accessible only to her account.
Goal	• Wants concrete, credible data to bring into a conversation with a specialist before a limited appointment slot.

(Scenario D) : FR-3,15,16,17,18,19,24

Mol has an upcoming appointment with a pediatric specialist for her 13-year-old son and wants concrete data to bring into the limited consultation time. Between shifts, she opens GrowTH, logs into her account (FR-3), selects her son's profile, and uploads a hand-and-wrist X-ray image (FR-15). The app validates the file format and size (FR-16), then submits it for processing.

After a short wait, the predicted bone age result appears (FR-17), along with an explanation of its margin of error and a clear note that it supports, rather than replace, clinical assessment (FR-18). The result is automatically saved to her son's profile, so she can review it alongside his growth and screening history later (FR-19).

Reassured that the image and result are only accessible through her own account (FR-24), Mon logs out (FR-3) closes the app confident she has something concrete to discuss at the upcoming appointment.

Persona 4 : Teacher Oi (Secondary user)

Photo	
Name	ครูอ้อย (Teacher Oi)
Age	45
Role	Secondary Persona / Secondary user (uses educational content but not child-tracking features)
Occupation	Elementary / Middle School Teacher (ครูโรงเรียนประถมหรือมัธยมต้น)

Background	Is a teacher or a general user who can access the application's educational and promotional materials and genuinely cares about the well-being and development of their students.
Motivation	- Care for Well-being: Driven to ensure students develop and grow properly for their age. - Need for Simple Media: Seeking jargon-free medical content and promotional videos that are easy to understand and recommend to parents. - Advocating Screening: Aiming to act as a bridge for parents to spot early warning signs (e.g., precocious puberty) and seek timely medical care.
Expectation	Expects to find easy-to-understand content or promotional videos that can be accurately shared and recommended to parents.
Goal	To access reliable child-development information and recommend useful tools and resources to parents when necessary.
Design Impact	Videos and promotional content should use simple, easy-to-understand language and avoid excessive medical jargon or parent-specific terminology, so that teachers and the general public can easily understand and recommend the application to others.

Scenario : Teacher Oi notices that one of her fifth-grade students has been growing unusually fast and has recently complained about bone pain. She becomes concerned that the child may be experiencing precocious puberty. To find reliable information, she accesses the GrowTH application as a general user and visits the knowledge center, where she reads articles about puberty and adolescent development.

She appreciates that the content is written in simple, easy-to-understand language without complicated medical terminology, making it accessible for parents. Teacher Oi then shares the article link and a promotional video about the application with the student's mother through LINE. She also recommends that the mother try the app's precocious puberty screening tool and AI-based bone age analysis feature. By doing so, the family can seek professional medical advice promptly if necessary, helping ensure that the child receives appropriate care at the right time.